ICML 2024 | Why Jacobian Regularisation Makes Networks Both Accurate and Robust

KAIST extends infinite-width theory to a network's input-output Jacobian, and shows that penalising it aligns accuracy with robustness instead of trading them off.

Over the last few years, infinite-width theory has given us remarkably clean answers about deep networks. Tools like the Neural Tangent Kernel (NTK) describe how a wide network initialises, how gradient descent moves its parameters, and where training ends up. But there is a quiet limitation running through all of it: this theory describes a network's output. It says almost nothing about the network's input-output Jacobian, the object that captures how the output changes as you nudge the input, and therefore how smooth the network is.

That gap matters because smoothness is exactly what we reach for when we want robustness. Jacobian regularisation, adding a penalty on the network's input-output Jacobian at each training point, is one of those methods that simply works: it is a simple and strong defence against adversarial perturbations. The awkward part is that nobody could really explain why. The infinite-width lens that cracked open ordinary training had never been pushed to cover the Jacobian, let alone training that deliberately penalises it.

This ICML 2024 paper by Taeyoung Kim and Hongseok Yang (KAIST) closes that gap. They prove that a multilayer perceptron (MLP) and its Jacobian jointly converge to a Gaussian process as width grows, define a Jacobian Neural Tangent Kernel (JNTK) that governs training, and show that Jacobian-regularised training, which they call robust training, behaves in the infinite-width limit exactly like a fixed kernel regression. Along the way they give the first infinite-width explanation of why penalising the Jacobian yields networks that are both accurate and robust.

Paper: https://arxiv.org/abs/2312.03386

The setup: from the output to the Jacobian

A short recap fixes the vocabulary. Infinite-width theory studies two limiting kernels. The NNGP kernel describes the distribution of a randomly initialised network's output, which converges to a Gaussian process as width grows. The NTK describes how that output moves under gradient descent, and its key virtue is that in the wide limit it becomes deterministic at initialisation and stays constant during training, which turns learning into a tractable linear system. Everything here concerns the scalar (or vector) output of the network.

Robust training changes the objective. On top of the usual mean-squared error, it adds a coefficient lambda times a penalty on the squared entries of the network's input-output Jacobian, evaluated at each training point. Larger lambda pushes the network toward a flatter input-output map. The question the paper asks is whether the same infinite-width machinery, GP convergence at initialisation and a constant kernel during training, can be extended from the output to this joint object of the network and its Jacobian.

Four theoretical moves

The analysis proceeds in four steps. First, the authors prove that at initialisation an MLP and its Jacobian jointly converge to a zero-mean Gaussian process, and they write down the limiting covariance, the Jacobian NNGP kernel, inductively over depth. The proof recasts this joint convergence as a tensor program and applies the Master theorem of the Tensor-Program framework, which is what lets them handle the network and its derivative together rather than one at a time.

Second, they define the Jacobian NTK (JNTK) from the inner products of the parameter-gradients of both the output and each Jacobian component, and prove the wide-limit analogue of the NTK's two defining properties: the JNTK becomes deterministic at initialisation, and it stays constant throughout robust training. Because bounding how far the parameters move now has to control the Jacobian penalty as well, the proof needs the 1-, 2-, and infinity-norms of the parameter movement, where the standard NTK argument gets by with the 2-norm alone.

Third, a constant kernel collapses the training dynamics. The learning trajectory reduces to a linear first-order ordinary differential equation (ODE), whose infinite-time solution is a plain kernel regressor built from the limiting JNTK, rescaled by a simple diagonal matrix that carries the regularisation strength lambda. In other words, robust training of a wide MLP is, in the limit, kernel regression with a Jacobian NTK. The fourth step is empirical: validate the convergence claims, then use the closed-form solution to study when accuracy and robustness go together.

It helps to see the pieces written out. Robust training minimises the mean-squared error plus lambda times the summed squares of the Jacobian entries at each training point. Once the JNTK is shown to be constant, the infinite-time solution takes the closed form f_ntk(x) = Theta_lambda(x, X)^T Theta_lambda(X, X)^(-1) y, an ordinary kernel regressor over the training inputs X. Its kernel Theta_lambda = Lambda Theta Lambda is nothing more than the limiting JNTK Theta wrapped by a diagonal matrix Lambda whose first entry is 1 and whose remaining entries are the square root of lambda. The regularisation strength therefore enters the solution only through this rescaling, which is what makes the effect of lambda so easy to reason about, and what the rest of the paper relies on when it compares robust training against standard training on an equal footing.

Does the theory hold up numerically?

The first experiments test the initialisation claim: as an MLP gets wider, its finite JNTK should close in on the deterministic limit. The authors use a controlled synthetic dataset, 256 points in four dimensions approximating an even covering of the sphere, and sweep every width from 64 up to 8192 (powers of two, 2^6 to 2^13) at depths 1, 2, and 3. Every configuration is repeated 10 times and reported with 95% bootstrap confidence intervals, and the covariance estimates rest on one million Monte-Carlo samples. Figure 1 tracks the max-norm distance between the scaled finite JNTK and its limit: for each depth the gap shrinks steadily as width grows, matching the prediction that the JNTK is deterministic at initialisation.

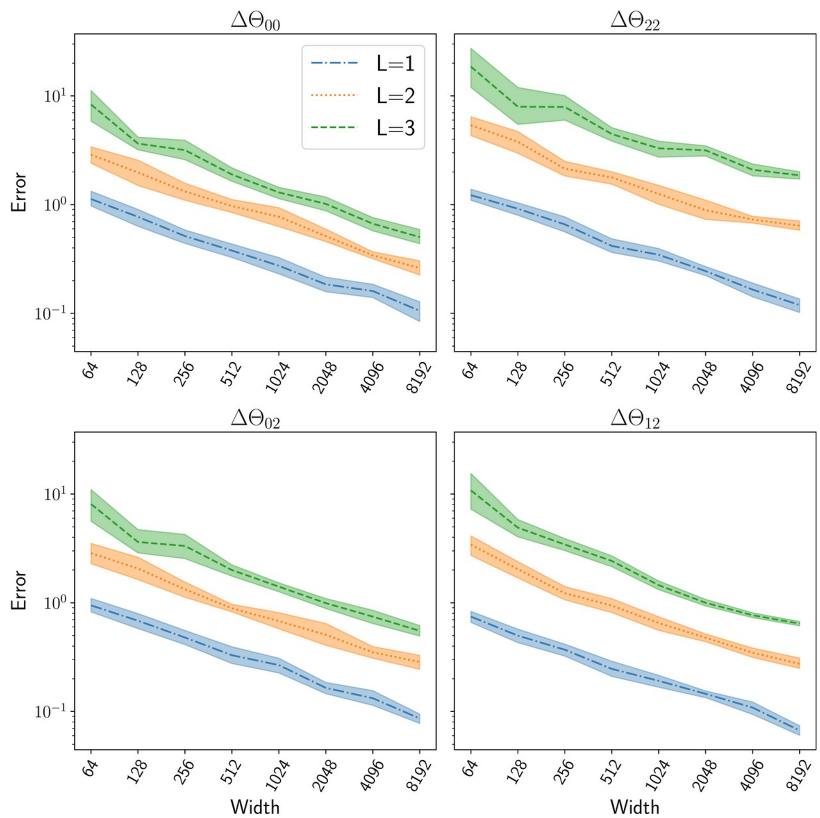

Figure 1. Max-norm distance between the (1/kappa^2)-scaled finite JNTK at initialisation and its limiting counterpart, across four representative kernel entries. The x-axis is MLP width (64 to 8192), the y-axis the error; colours mark depth L = 1, 2, 3. Within each depth the error falls as the network widens, supporting the claim that the JNTK becomes deterministic in the infinite-width limit.

The second experiment is the more demanding one: the JNTK should not just be deterministic at initialisation, it should stay put during robust training. Here the authors switch to the Algerian forest fire dataset (UCI, 224 points, 11 input features) treated as a plus-or-minus-one regression task, and track the kernel's deviation from its initialisation-time limit at each training step, again sweeping the same range of widths. This is a strictly harder claim than the initialisation result, because it has to hold not just at one moment but all along the optimisation trajectory as the network trains.

Robust training here uses a Jacobian coefficient lambda = 0.01, a scaling constant kappa = 0.1, and a learning rate of 1, and the kernel is measured at a sequence of training steps t running from 0 up to 2048. Figure 2 shows the payoff: at every training step checked, the deviation from the limit shrinks monotonically as width grows. In other words, the wider the network, the more faithfully the JNTK holds its initial value throughout training, exactly as the constancy theorem predicts it should.

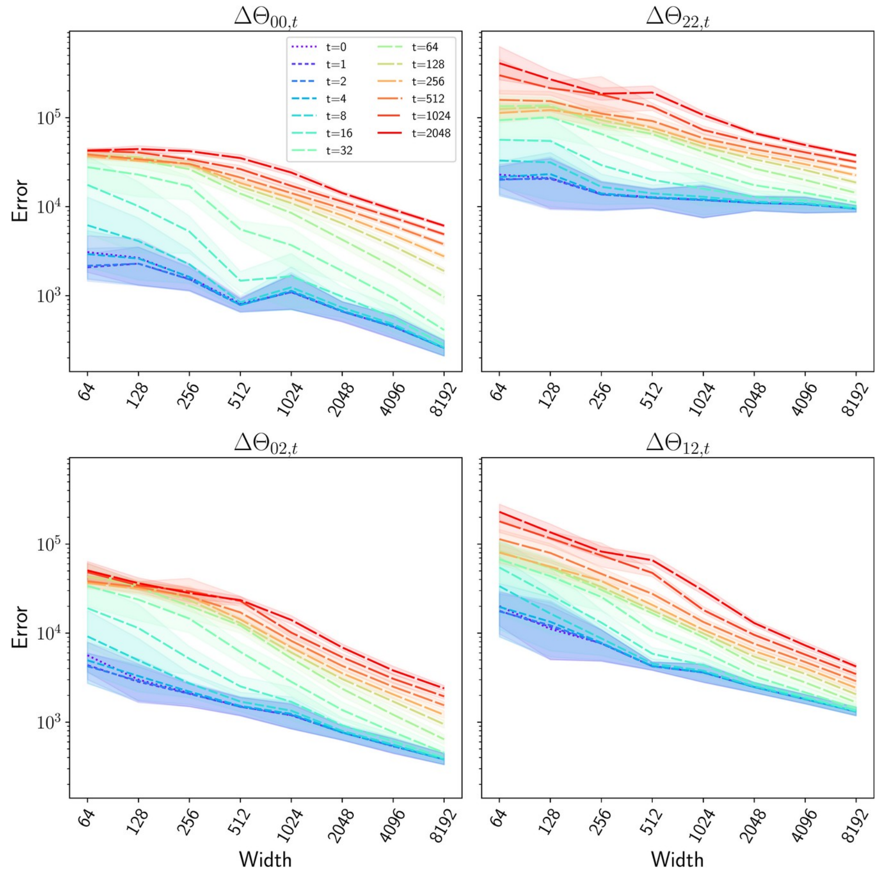

Figure 2. Deviation of the scaled finite JNTK from its initialisation-time limit during robust training, plotted against width for a range of training steps t (colours, t = 0 to 2048). Across the whole sweep the error decreases with width at every step, evidence that the JNTK stays approximately constant throughout robust training in the wide limit.

The authors are candid about the boundary of this validation. At the widest network tested (8192), the max-norm error is still larger than the smallest eigenvalue of the relevant Gram matrix, so the experiments do not fully certify the bound they would need for an exact guarantee. But the trend is unambiguous, wider is closer at every step, and fully closing the gap would simply require wider networks than the compute budget allowed.

The payoff: accuracy and robustness stop competing

With a closed-form kernel-regression solution in hand, the authors can inspect its eigenfeatures directly, asking whether the features that make the solution accurate are the same ones that make it robust. Robustness here is measured by how test accuracy holds up after small perturbations (0.01 and 0.1) of the eigenfeatures. The striking finding is that under Jacobian regularisation, accuracy and robustness move together: the more accurate eigenfeatures are also the more robust ones.

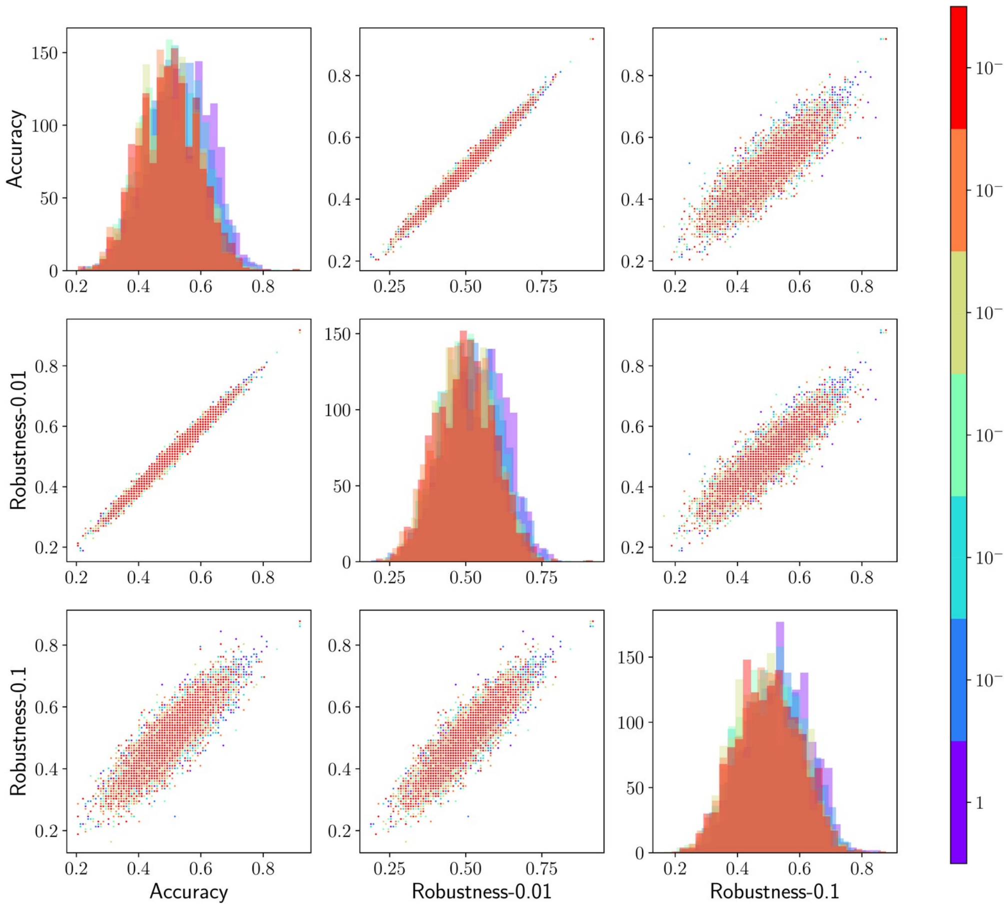

Figure 3. Pairplot of test accuracy against accuracy after 0.01 and 0.1 perturbations, for eigenfeatures of the robust-training (Jacobian-regularised) solution; colour encodes the regularisation coefficient. The point clouds run along the diagonal, so more accurate eigenfeatures are also more robust, accuracy and robustness align rather than trade off.

Standard training tells the opposite story. Running the same analysis on the ordinary (non-regularised) solution, Figure 4 shows little to no correlation between accuracy and robustness. Just as important, standard training never produces the highly-accurate-but-fragile eigenfeatures reported in earlier finite-NTK work by Tsilivis and Kempe (2022). The comparison suggests that the Jacobian regulariser is not merely trading a little accuracy for robustness; it reshapes the solution so the two properties reinforce each other.

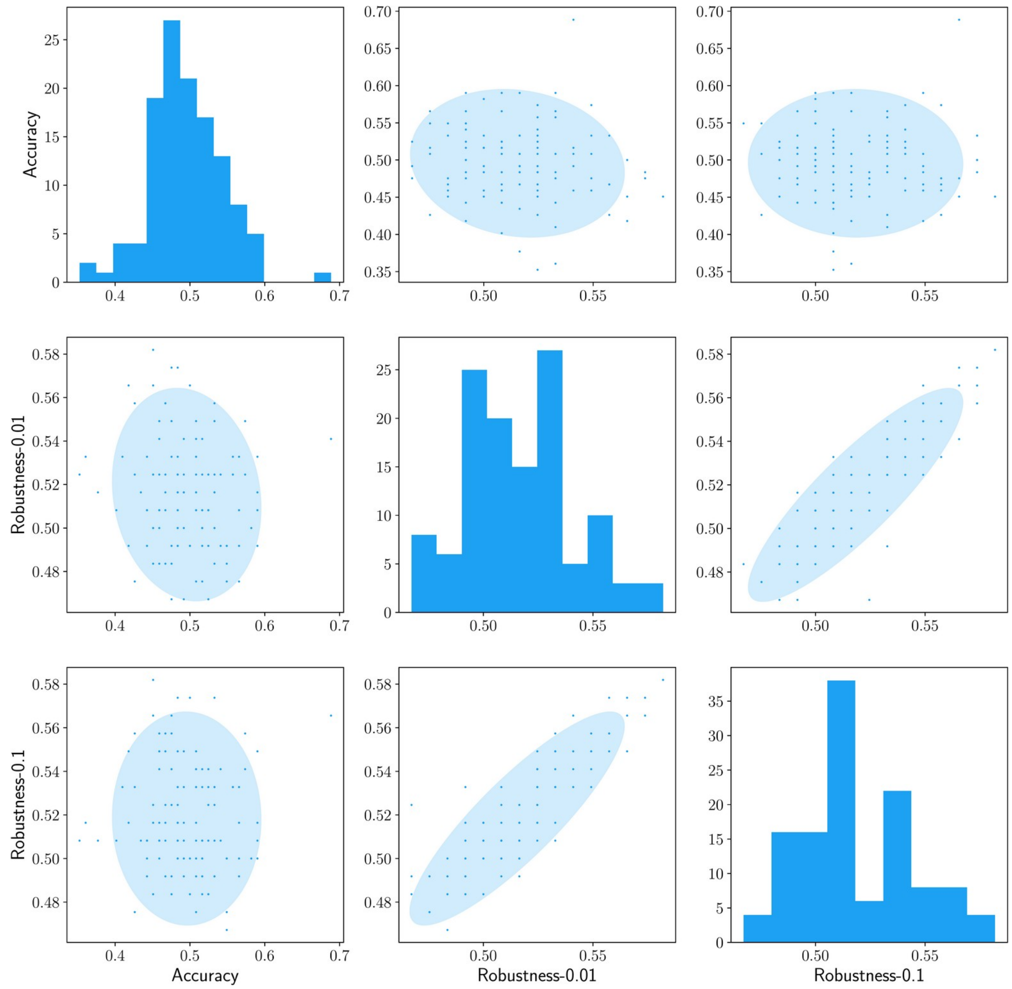

Figure 4. The same pairplot for standard (non-regularised) training; ellipses mark the 4-sigma confidence region of a Gaussian fit to make the correlation legible. Accuracy and robustness show little correlation, and no accurate-but-fragile eigenfeatures appear, in contrast to the robust-training solution in Figure 3.

A caveat: the full-rank assumption is fragile

The clean kernel-regression story rests on an assumption (the paper's Assumption 4.4): the limiting JNTK must have a nonzero smallest eigenvalue, i.e. its Gram matrix must be full-rank. The authors check when this actually holds, and the answer is a useful practical warning. Figure 5 plots the smallest eigenvalue of the JNTK and the standard NTK against depth for two activations. The JNTK's smallest eigenvalue is far below the NTK's, and it only becomes positive for deeper networks: depth 11 or more for GeLU, but as shallow as depth 6 for the erf activation.

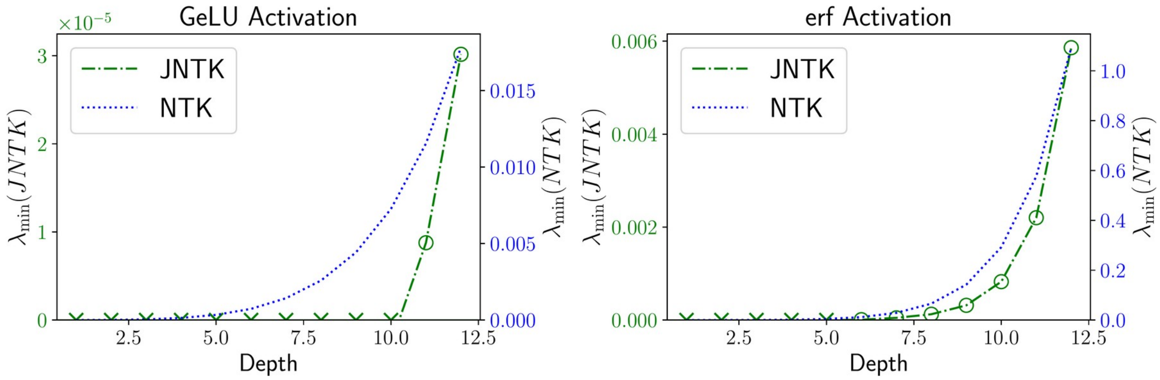

Figure 5. Smallest eigenvalue of the JNTK (green) and the standard NTK (blue) versus network depth, for GeLU (left) and erf (right) activations. Circles mark depths where the nonzero-minimum-eigenvalue assumption holds, crosses where it is violated. The JNTK needs depth >= 11 with GeLU but only >= 6 with erf, so the assumption behind the theory is far from automatic.

This is a genuinely useful diagnostic rather than a footnote: it tells a practitioner that the infinite-width guarantees for Jacobian-regularised training are activation- and depth-dependent, and gives a concrete rule of thumb for when the underlying condition can be expected to hold.

Takeaway

The bottom line is that infinite-width theory now reaches a network's Jacobian, not just its output. Train a wide MLP with a Jacobian regulariser and, in the limit, it behaves like kernel regression with a Jacobian Neural Tangent Kernel, and that same regulariser makes accuracy and robustness pull in the same direction instead of fighting each other. It is the first principled explanation, from the infinite-width viewpoint, of why penalising the Jacobian produces networks that are both accurate and robust, and it comes with a practical test (via the smallest eigenvalue) for when its key assumption holds.

The honest boundaries are worth keeping in view. The experiments are deliberately small and controlled, MLPs on synthetic and low-dimensional datasets, and full numerical certification of the constancy bound would need wider networks than were tested. But as a piece of theory, the work extends a mature toolset to a new object and turns a well-known empirical trick into something we can reason about.